Anway S. Pimpalkar

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EDUCATION

Harvard University, Cambridge, MA — 2025 –
PhD, Electrical Engineering
Advisor: Shriya S. Srinivasan, PhD

Johns Hopkins University, Baltimore, MD — **GPA:** 4.00 / 4.00 — 2023 – 2025
MSE, Biomedical Engineering (Neuroengineering Focus)
Thesis: Bridging Perception and Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Stimuli
Advisor: Jeremy D. Brown, PhD

College of Engineering Pune, Pune, India — **GPA:** 8.24 / 10.00 — 2019 – 2023
BTech, Electronics and Telecommunication Engineering

AWARDS AND HONORS

- **Harvard SEAS Prize Fellowship** ($281,796 • 6-yr tuition & 2-yr stipend) • Harvard University — 2025 –
- **National Defense Science & Engineering Graduate Fellowship** ($277,484 • 3-yr tuition & stipend) • DOD — 2025 –
- **GEM Associate Fellow** • The National GEM Consortium — 2025 –
- **Best Student Paper Finalist** • International Conference on Rehabilitation Robotics (ICORR / RehabWeek) — 2025
- **National Institute on Aging Start-Up Challenge Finalist** ($10,000) • NIH — 2025
- **Best Paper Award** • International Conference on Biomedical Engineering Science and Technology, India — 2023
- **Mitacs Globalink Research Award** ($9,915) • Dalhousie University, Canada — 2022
- **Best Project Award** • IEEE National Project Competition, India — 2021

RESEARCH EXPERIENCE

Johns Hopkins University • Haptics and Medical Robotics Lab ↗ — Baltimore, MD

Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD ↗ — Jan 2024 – Jun 2025

Multimodal Visual-Haptic Neurorehabilitation for Post-Stroke Finger Dexterity
Developed and evaluated a multimodal feedback framework for motor rehabilitation. Integrated cross-modal perceptual calibration to optimize sensory feedback delivery to individual thresholds. Implemented on ubiquitous devices to support smartphone-based rehabilitation outside the lab. *Publications:* P4, C2 (Best Student Paper Finalist at ICORR 2025), C3, A7

Pneumatactors: Soft Multimodal Haptic Interfaces
Engineered "pneumatactors" – a novel <u>pneumat</u>ic <u>tactor</u> haptic interface designed for co-located stimulation of fast and slow adapting mechanoreceptors, enhancing cognitive salience. 3D-printed with flexible TPE to match mechanical impedance of human skin for intuitive interaction. *Publications:* P5, A4, A5

Harvard University • Biohybrid Organs and Neuroprosthetics Lab ↗ — Cambridge, MA

Research Fellow • Advisor: Shriya S. Srinivasan, PhD ↗ — May 2024 – Aug 2024

Peripheral Nerve Stimulation to Modulate Cancer Tumor Metastasis
Developed non-invasive peripheral nerve stimulation strategies to modulate tumor growth and metastasis in murine models, revealing frequency-dependent effects on tumor innervation. Designed and conducted pilot studies on melanoma-bearing mice, validating outcomes through immunohistochemical analysis.

Surgical Approach for Treating Stress Urinary Incontinence
Engineered miniaturized, RF-powered vagus nerve stimulators for freely moving rodents to study glioblastoma–nervous system interactions. Devices delivered monophasic pulses up to 1 kHz via custom PCB oscillators, enabling targeted parasympathetic modulation in vivo.

IIT Bombay • Human Motor Neurophysiology and Neuromodulation Lab ↗ — Mumbai, India

Research Intern • Advisor: Nivethida T, PhD ↗ — Jan 2023 – Aug 2023

Quantifying Prospective Component of Sense of Agency using Machine Learning

Characterized pre-movement EEG signatures underlying the Sense of Agency using a modified Libet clock task with probabilistic auditory feedback. Applied spectral, time-series, and machine learning analyses to identify spatiotemporal patterns predictive of intentional binding. Demonstrated distinct neural correlates in pre-SMA and centroparietal regions, supporting a prospective model of agency. *Publications:* A2, Bachelor's Dissertation

College of Engineering Pune • Departments of Electronics, Applied Sciences ↗ Pune, India

Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD ↗ Aug 2022 – Aug 2023

Deep Learning-Based MRI Skull Tissue Segmentation

Developed and benchmarked vanilla, residual, and dense 2D U-Net architectures for skull stripping in neuroimaging, achieving 99.75% accuracy with a dense model optimized for feature reuse. Implemented data augmentation to improve generalizability across multi-scanner datasets. Demonstrated utility of open deep learning pipelines for replacing proprietary tools in MRI workflows. *Publications:* C1 (Best Paper Award at ICBEST 2023), A1

Dalhousie University • Agricultural Mechanized Systems Group ↗ Truro, Canada

Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD ↗ May 2022 – Aug 2022

Automating Harvesters to Reduce Operator Cognitive Load

Engineered a sensor-driven control system to automate bin switching and dynamic harvester head height adjustment, reducing operator load. Used RGB and ToF cameras for volumetric bin fill estimation (±10% accuracy), and integrated crop height and GPS data via CAN bus to actuate harvesting arms in real time. System validated over two harvest days with positive field performance.

Queliz Lifetech • Early Stage Medtech Startup ↗ Pune, India

Research Intern • Electronics Subsystem Jun 2021 – Sep 2021

Point-of-Care Device for Finger Dexterity Rehabilitation in Acute Burn Patients

Designed an affordable, portable rehabilitation device to restore MCP, DIP, and PIP joint mobility in burn patients. Developed a precision-controlled actuation system with adaptive feedback to enable customized therapy in low-resource clinical settings.

JOURNAL PUBLICATIONS

P5 **A. S. Pimpalkar** et al., "Pneumatactors: Soft 3D-printed interface for co-located transient and continuous tactile stimuli," in preparation, 2025.

P4 **A. S. Pimpalkar**, A. M. West, D. Rai, J. Xu, and J. D. Brown, "Evaluation of multimodal feedback to augment motor control in precision grip," in preparation, 2025.

P3 C. Harris, **A. S. Pimpalkar**, A. Aggarwal, J. Yang, X. Chen, S. Schmidgall, S. Rapuri, J. Greenstein, C. Overby-Taylor, and R. D. Stevens, "Risk prediction for non-cardiac surgery using the 12-lead electrocardiogram: An explainable deep learning approach," *British Journal of Anesthesiology* (in print), 2025. Preprint: medRxiv ↗

P2 **A. S. Pimpalkar**, A. Slepyan, and N. V. Thakor, "Vibrations at first contact encode object stiffness before grasp completion," *IEEE Sensors Letters*, June 2025. doi: 10.1109/LSENS.2025.3584860 ↗

In Media: *Featured in New Scientist Magazine.* Article ↗

P1 **A. S. Pimpalkar** and D. Niture, "Towards contactless elevators with tinyML using person detection and keyword spotting," Jul 2024. Best Project Award. Preprint: arXiv ↗, Code ↗

CONFERENCE PRESENTATIONS † presenter

FULL-LENGTH PAPERS

C3 A. M. West, **A. S. Pimpalkar**, J. Xu, and J. D. Brown, "SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments," *International Conference on Neural Engineering*, June 2025 (in review).

C2 **A. S. Pimpalkar** †, A. M. West, J. Xu, and J. D. Brown, "Optimizing cross-modal matching for multimodal motor rehabilitation," *International Conference on Rehabilitation Robotics (ICORR / RehabWeek)*, May 2025. Best Student Paper Award Finalist. *Podium presentation.*

C1 **A. S. Pimpalkar** †, R. K. Patole, K. D. Kamble, and M. H. Shindikar, "Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans," *International Conference on Biomedical Engineering Sci. & Tech.*, Feb 2023. Best Paper Award. *Podium presentation.* Paper ↗

ABSTRACTS AND SHORT ARTICLES

A7 **A. S. Pimpalkar** †, D. Rai, J. U. Bartels, J. Xu, and J. D. Brown, "Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task," *Biomedical Engineering Society (BMES),* Oct 2024. *Podium pres.* Abstract ↗

A6 H. Song † et al. [including **A. S. Pimpalkar**], "Towards visual function restoration through photoacoustic stimulation," *Association for Research in Vision & Ophthalmology (ARVO)*, May 2024. *Podium presentation.* Abstract ↗

A5 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia, and J. D. Brown, "Pneumatactor arrays for high frequency vibrotactile feedback," *IEEE Haptics Symposium*, Apr 2024. *Live demonstration and Work-in-Progress paper (2 presentations)*.

A4 **A. S. Pimpalkar** †, P. Ameta †, A. Dalia †, and J. D. Brown, "Pneumatactor arrays for high frequency vibrotactile feedback," *United States Senate AI Caucus: Robotics Demo Day*, Apr 2024. *Live demonstration.*

A3 C. Harris †, **A. S. Pimpalkar**, A. Aggarwal, P. Yang, X. Chen, C. Overby-Taylor, J. Greenstein, and R. D. Stevens, "Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms," *Johns Hopkins Medicine & Engineering Research Retreat*, Feb 2024. *Poster presentation.* Poster ↗

A2 **A. S. Pimpalkar** †, R. Patole, and N. Thirugnanasambandam, "Demonstrating the prospective component of sense of agency using machine learning," *COEP ENTC BTech Project Symposium* , May 2023. *Poster presentation.* Poster ↗

A1 **A. S. Pimpalkar** †, R. Patole, K. Kamble, and M. Shindikar, "Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans," *No Garland Neuroscience Conference*, Feb 2023. *Podium and poster presentation.* Poster ↗

TEACHING ENGAGEMENTS

Mechatronics, Johns Hopkins • Senior Level — Spring 2024

Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge in real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking (open-source implementation ↗). Led labs on building autonomous robots. Worked in a mentorship role in Spring 2025.

Principles of the Design of Biomedical Instrumentation, Johns Hopkins • Graduate Level — Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG recording circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction, Johns Hopkins • Graduate Level — Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Helped pilot an online learning platform to provide self-paced lectures incorporating live instructor interaction.

INVITED TALKS

Haptics: Neural Basis, Applications, Future Avenues. **Johns Hopkins University, Sep 2024**. **Neural & Rehabilitation Engineering Course**, invited by Nitish V. Thakor, PhD.

SKILLS AND CERTIFICATIONS

Medical Research	Human participant studies (non-invasive devices), rodent studies (invasive devices, mice/rats), human perception experiment design, IRB protocols
Electronics	Analog and digital circuit design, microcontroller programming (Arduino/Teensy/ESP/STM), high-speed and high-current PCB design (KiCAD/Altium), closed-loop control systems, DAQ design, FPGA, RF design, precision soldering (THT/SMT), mechatronic integration, communication protocols

Programming & CS	Python, MATLAB, C/C++, data analysis, VR rendering, machine and deep learning, TinyML, TensorFlow (+lite/micro), computer vision, LaTeX
Fabrication & Design	CAD (Autodesk, Shapr3D), FDM 3D printing (rigid/flexible), SLA resin printing, silicone molding
Certifications ↗ (15 total courses)	**Harvard Online** • TinyML Certificate (3 courses), Python for Research, CS50 Web Development **MIT Online** • Aeronautics and Human Spaceflight **Johns Hopkins Online** • Human Subject Research, Neuroimaging, Neurohacking **DeepLearning.AI** • Deep Learning Specialization (5 courses) **AICTE** • Computer Science & Biology (faculty development program completed with permission)

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer 2024
Canadian Society of Bioengineering Annual Meeting • Student Volunteer 2022

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Student Leader, Operations and Logistics 2024 – 2025

- Managed operations, logistics, sponsorships for university's first Hopkins India Conference, Washington DC.

Johns Hopkins Biomedical Engineering • Master's Program Representative 2024 – 2025

COEP Impressions Cultural Festival • Head of Events (2021), Volunteer 2019 – 2021

- Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinated with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements.
- Built and led a team of 17, working with 2 co-heads to build a platform from the ground up, fostering opportunities for budding artists across India and uniting them under a shared vision. Placed emphasis on unconventional art forms.

Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler 2020 – 2022

- Won multiple race events and an institutional award commending motivation and skill. Represented city at regional races.

COEP Mental Health & Wellness Center • Student Volunteer 2020 – 2022

COEP Data Science & AI Club • Research Lead (2020-21), Member 2020 – 2022